

Topic 4 Summary: Projectile Motion

- Introduction
- Definitions and derivations of equations for
 - Time of flight
 - Greatest height
 - Horizontal range/maximum range
- Types of projectile motion
 - Vertically launched
 - Horizontally launched
- Worked examples

Two-dimensional Motion

So far we have introduced the concepts of kinematics to describe motion in one-dimension; however we live in a multidimensional universe. In order to explore and describe motion in this universe, we begin by looking at examples of two-dimensional motion. Two common examples are planets orbiting a star in elliptical orbits or a projectile moving under the influence of gravity.

Projectile Motion

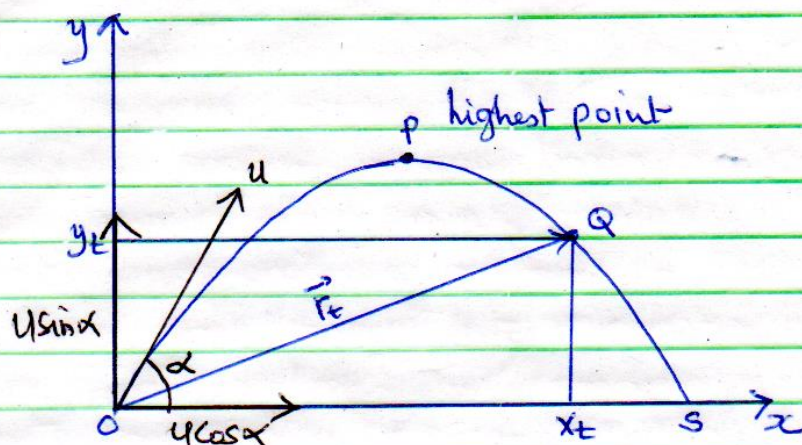
A projectile is a body moving freely under gravity. e.g a missile shot from a launcher, an object dropped from an aircraft or a football kicked.

In order to investigate the motion of a projectile, we should consider the horizontal (x) and vertical (y) motions separately.

Assuming negligible air-resistance, the horizontal motion is of constant velocity since there is no acceleration in this direction.

The vertical motion is subject to a constant downward gravitational acceleration, g , and the usual equations of motion: $v = u + at$, $v^2 = u^2 + 2ax$ and $x = ut + \frac{1}{2}at^2$ are used.

Consider an object which is thrown with an initial velocity u at an angle α to the horizontal, as shown in the figure below:



At any time t , we consider the x and y motions:

Horizontal motion (x)	Vertical motion (y)
$x = (u \cos \alpha) t$	$y = (u \sin \alpha) t - \frac{1}{2} g t^2$
$v_x = u \cos \alpha$	$v_y = u \sin \alpha - g t$
$\vec{a}_x = 0$	$a_y = -g$
$t = \text{time taken}$	$t = \text{time taken}$

Note:

- the horizontal motion is completely independent of the vertical motion.
- The path followed by a projectile is usually known as its trajectory.
- If we eliminate t in equations of x and y above, we get the equation of a parabola, which is the shape of this trajectory.

Time of flight

The time taken for the projectile to travel along its path from O to S , is known as the time of flight.

In this case, $y = 0$, therefore from above!

$$(u \sin \alpha)t - \frac{1}{2}gt^2 = 0$$

$$t(u \sin \alpha - \frac{1}{2}gt) = 0$$

$$\Rightarrow t = 0 \quad \text{or} \quad t = \frac{2u \sin \alpha}{g}$$

↓
object at 0

↓
object at s.

∴ time of flight is

$$\boxed{t = \frac{2u \sin \alpha}{g}} \quad \text{--- (2)}$$

Greatest Height

The highest point on the parabola is reached when the vertical component of velocity is zero. i.e. $V_y = 0$.

$$\therefore \text{ we have } 0 = u \sin \alpha - gt$$

$$\boxed{t = \frac{u \sin \alpha}{g}} \quad \text{--- (3)}$$

we notice that the time to reach the highest point is half the time of total flight. (because the parabola is symmetrical, time OP = time PS).

Then the greatest height reached, h , is :

$$\begin{aligned} h = y &= (u \sin \alpha)t - \frac{1}{2}gt^2 \\ &= u \sin \alpha \left(\frac{u \sin \alpha}{g} \right) - \frac{1}{2}g \left(\frac{u \sin \alpha}{g} \right)^2 \\ &= \frac{u^2 \sin^2 \alpha}{g} - \frac{1}{2} \frac{u^2 \sin^2 \alpha}{g} \end{aligned}$$

$$\therefore \boxed{h = \frac{u^2 \sin^2 \alpha}{2g}} \quad \text{--- (4)}$$

Note : h increases as u increases, also h inc as α inc.

Horizontal Range

The displacement OS in the horizontal direction is known as the horizontal range or simply the range of the projectile.

$$x = (u \cos \alpha) t \quad \text{but } t = \text{time of flight} = \frac{2u \sin \alpha}{g}$$

$$\therefore x = (u \cos \alpha) \left(\frac{2u \sin \alpha}{g} \right)$$

$$x = OS = \frac{2u^2 \sin \alpha \cos \alpha}{g} = \boxed{\frac{u^2 \sin 2\alpha}{g}} \quad - (5)$$

Maximum Range

For a given value of u , the horizontal range $\frac{u^2 \sin 2\alpha}{g}$, would have the maximum value

when $\sin 2\alpha = 1$. This occurs when $\alpha = 45^\circ$.

Thus the maximum horizontal range is $\boxed{\frac{u^2}{g}} - (6)$

and it occurs ~~at~~ when the angle of projection is 45° .

Example 1.

A particle is projected from a point on a horizontal plane and has an initial velocity of $28\sqrt{3}$ m/s at an angle of elevation of 60° . Find the greatest height reached by the particle and the time taken to reach that height.

Solution

$$H = \frac{u^2 \sin^2 \alpha}{2g} = \frac{(28\sqrt{3})^2 (\sin 60^\circ)^2}{2 \times 9.8}$$

$$H = \underline{90\text{m}}$$

Time to reach greatest height, $t =$

$$t = \frac{u \sin \alpha}{g}$$

$$= \frac{28\sqrt{3} \cos 60^\circ}{9.8} = \underline{\underline{4.29 \text{ sec}}}$$

Example 2

A particle is projected from an origin O with a velocity of $(30\hat{i} + 40\hat{j})$ m/s.

a) Find the position and velocity vectors of the particle 5 seconds later.

b) Hence find the distance of the particle from O and its speed and direction of motion at this time. (Take $g = 10 \text{ m/s}^2$).

Solution.

$$\text{a) } u = \begin{pmatrix} 30 \\ 40 \end{pmatrix} \text{ m/s.}$$

$$\text{Using } S = ut + \frac{1}{2}at^2.$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 30 \\ 40 \end{pmatrix}t + \frac{1}{2} \begin{pmatrix} 0 \\ -10 \end{pmatrix}t^2, \text{ but } t=5$$

$$\therefore \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 30 \\ 40 \end{pmatrix}5 + \frac{1}{2} \begin{pmatrix} 0 \\ -10 \end{pmatrix}5^2 = \begin{pmatrix} 150 \\ 200 \end{pmatrix} + \begin{pmatrix} 0 \\ -125 \end{pmatrix} = \begin{pmatrix} 150 \\ 75 \end{pmatrix}$$

$\therefore$ position vector is $150\hat{i} + 75\hat{j}$

Similarly, for velocity vector

$$v = u + at$$

$$\begin{pmatrix} v_x \\ v_y \end{pmatrix} = \begin{pmatrix} 30 \\ 40 \end{pmatrix} + \begin{pmatrix} 0 \\ -10 \end{pmatrix} 5$$
$$= \begin{pmatrix} 30 \\ 40 \end{pmatrix} + \begin{pmatrix} 0 \\ -50 \end{pmatrix} = \begin{pmatrix} 30 \\ -10 \end{pmatrix}$$
$$\vec{v} = \underline{30\hat{i} - 10\hat{j}}$$

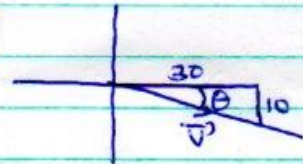
$$\text{b) Distance} = |\text{displacement}|$$
$$= \sqrt{150^2 + 75^2} = 75\sqrt{5} \text{ m} = 167.7 \text{ m}$$

$$\text{Speed} = |v| = 10\sqrt{10} = 31.62 \text{ m/s}$$

direction of motion = direction of $\vec{v}$

$$\therefore \tan \theta = \frac{10}{30} \Rightarrow \theta = 18.43^\circ$$

$\therefore$ direction of motion is at a bearing of 108.43°



Example 4.

A stone is thrown horizontally over the edge of a vertical cliff with an initial speed of 20 m/s. The cliff is 150 m high.

- Find the distance from the cliff at which the stone hits the water.
- Find also the magnitude and direction of the velocity at this instant.

Solution.

$$(i) \quad u = \begin{pmatrix} 20 \\ 0 \end{pmatrix} \quad s = \begin{pmatrix} R \\ -150 \end{pmatrix} \quad a = \begin{pmatrix} 0 \\ -10 \end{pmatrix} \quad t = t$$

$$s = ut + \frac{1}{2}at^2$$

$$\begin{pmatrix} R \\ -150 \end{pmatrix} = \begin{pmatrix} 20 \\ 0 \end{pmatrix}t + \frac{1}{2}\begin{pmatrix} 0 \\ -10 \end{pmatrix}t^2$$

$$R = 20t \quad \text{--- (i)}$$

$$-150 = -5t^2 \quad \text{--- (ii)}$$

from (ii)

$$t = \sqrt{30} = 5.477 \text{ sec}$$

∴ putting this value of t in (i)

$$R = 20(5.477)$$

$$= \underline{\underline{109.54 \text{ m}}}$$

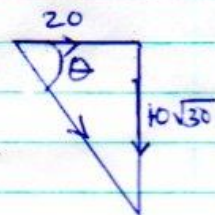
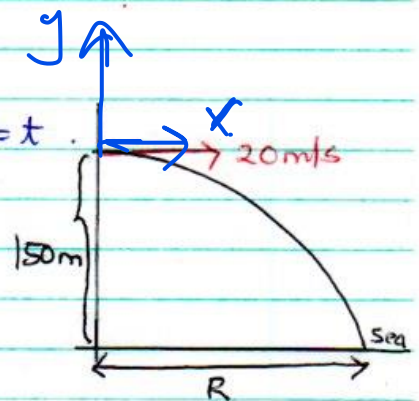
$$(ii) \quad v = u + at$$

$$= \begin{pmatrix} 20 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ -10 \end{pmatrix}\sqrt{30} = \begin{pmatrix} 20 \\ -10\sqrt{30} \end{pmatrix} \text{ m/s}$$

$$|v| = \sqrt{20^2 + (-10\sqrt{30})^2} = 58.3 \text{ m/s}$$

$$\tan \theta = \frac{-10\sqrt{30}}{20}$$

$$\theta = 69.9^\circ$$



Example 5:

Find the initial angle of elevation of a projectile whose horizontal range is 5 times its greatest height.

Sol:

$$\frac{u^2 \sin 2\alpha}{g} = \frac{5u^2 \sin^2 \alpha}{2g}$$

$$2 \sin \alpha \cos \alpha = \frac{5}{2} \sin \alpha \sin \alpha$$

Divide both sides by $\sin \alpha \cos \alpha$

$$2 = \frac{5}{2} \tan \alpha$$

$$\tan \alpha = \frac{4}{5} \Rightarrow \alpha = \underline{\underline{38.66^\circ}}$$

Example 6:

The maximum distance a boy can throw a ball is 40m. At what angle of elevation would he achieve a distance of 25m, throwing the ball at the same speed?

Sol:

$$\frac{u^2}{g} = 40 \Rightarrow u^2 = 400$$
$$\therefore u = 20 \text{ m/s.}$$

$$R = \frac{u^2 \sin 2\alpha}{g}$$

$$25 = \frac{400 \sin 2\alpha}{10} \Rightarrow \sin 2\alpha = \frac{25}{40}$$

$$2\alpha = 38.6^\circ$$

$$\therefore \alpha = \underline{\underline{19.34^\circ}}$$

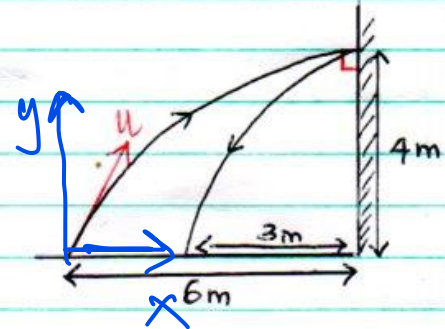
Example 7

A ball is thrown from a point on the ground towards a vertical wall which is 6m away. It strikes the wall at right angles at a height of 4m and bounces back to hit the ground at a point 3m from the wall. Find the ball's initial elevation and speed and the coefficient of restitution for the ball and the wall.

Solution.

$$R = 12\text{m and } H = 4\text{m}$$

(Since it hits at right angles,
 $\Rightarrow$ 4m is max height)
 $\therefore$ 6m is half range)



$$\therefore R = 3H.$$

$$\frac{u^2 \sin 2\alpha}{g} = \frac{3u^2 \sin^2 \alpha}{2g}$$

$$2 \sin \alpha \cos \alpha = \frac{3 \sin^2 \alpha}{2}$$

$$\tan \alpha = \frac{4}{3}$$

$$\alpha = 53.13^\circ$$

$$\text{If } \tan \alpha = 4/3 \quad \begin{array}{c} 5 \\ \alpha \\ 3 \end{array} \quad 4 \Rightarrow \sin \alpha = 4/5 \quad \cos \alpha = 3/5$$

$$R = 12 = \frac{u^2 \sin 2\alpha}{g} = \frac{2u^2 \sin \alpha \cos \alpha}{g}$$

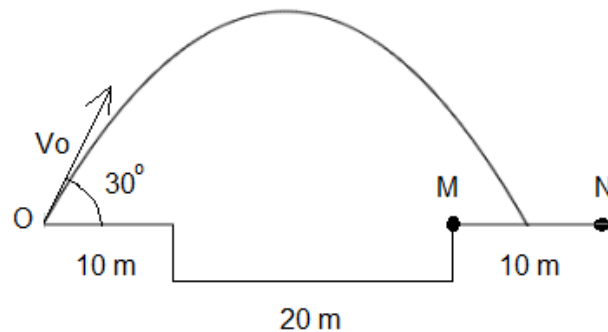
$$12 = \frac{2u^2 (4/5)(3/5)}{10}$$

$$u = \underline{\underline{11.2 \text{ m/s}}}$$

Example 8

A projectile is to be launched at an angle of 30° so that it falls beyond the pond of length 20 meters as shown in the figure.

What is the range of values of the initial velocity so that the projectile falls between points M and N?

**Solution**

The range is given by $x = V_0^2 \sin(2\theta) / g$

We want to have the range greater than OM and smaller than ON, with $OM = 10 +$

$20 = 30$ m and $ON = 10 + 20 + 10 = 40$ m

$$30 < V_0^2 \sin(2\theta) / g < 40$$

$$30 g / \sin(2\theta) < V_0^2 < 40 g / \sin(2\theta)$$

$$\sqrt{[30 g / \sin(2\theta)]} < V_0 < \sqrt{[40 g / \sin(2\theta)]}$$

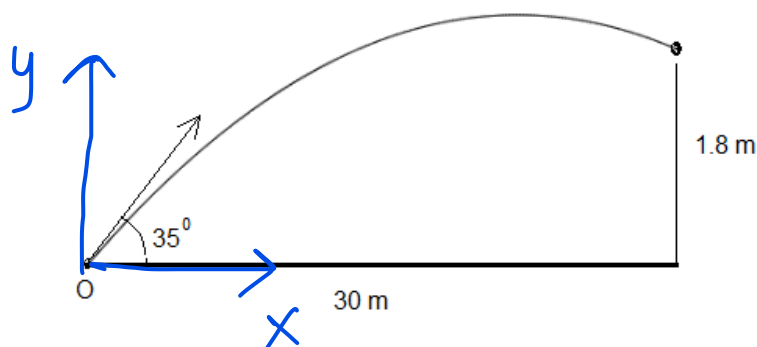
$$18.4 \text{ m/s} < V_0 < 21.2 \text{ m/s}$$

Example 9

A ball is kicked at an angle of 35° with the ground.

a) What should be the initial velocity of the ball so that it hits a target that is 30 meters away at a height of 1.8 meters?

b) What is the time for the ball to reach the target?



Solution

$$\text{a) } x = V_0 \cos(35^\circ) t$$

$$30 = V_0 \cos(35^\circ) t$$

$$t = 30 / V_0 \cos(35^\circ)$$

$$1.8 = -(1/2) 9.8 (30 / V_0 \cos(35^\circ))^2 + V_0 \sin(35^\circ)(30 / V_0 \cos(35^\circ))$$

$$V_0 \cos(35^\circ) = 30 \sqrt{[9.8 / 2(30 \tan(35^\circ) - 1.8)]}$$

$$V_0 = 18.3 \text{ m/s}$$

$$\text{b) } t = x / V_0 \cos(35^\circ) = 2.0 \text{ s}$$

Example 10

A ball kicked from ground level at an initial velocity of 60 m/s and an angle θ with ground reaches a horizontal distance of 200 meters.

a) What is the size of angle θ ?

b) What is time of flight of the ball?

Solution

a) Let T be the time of flight. Two ways to find the time of flight

1) $T = 200 / V_0 \cos(\theta)$ (range divided by the horizontal component of the velocity)

2) $T = 2 V_0 \sin(\theta) / g$ (formula found in projectile equations)

equate the two expressions

$$200 / V_0 \cos(\theta) = 2 V_0 \sin(\theta) / g$$

which gives

$$2 V_0^2 \cos(\theta) \sin(\theta) = 200 g$$

$$V_0^2 \sin(2\theta) = 200 g$$

$$\sin(2\theta) = 200 g / V_0^2 = 200 (9.8) / 60^2$$

Solve for θ to obtain

$$\theta = 16.5^\circ$$

$$\text{b) Time of flight} = 200 / V_0 \cos(16.5^\circ) = 3.48 \text{ s}$$

Example 11

The trajectory of a projectile launched from ground is given by the equation $y = -0.025x^2 + 0.5x$, where x and y are the coordinates of the projectile on a rectangular system of axes.

Find the initial velocity and the angle at which the projectile is launched.

Solution

$$y = \tan(\theta)x - (1/2)(g / (V_0 \cos(\theta))^2)x^2 \text{ (formula found in projectile equations)}$$

$$\text{hence } \tan(\theta) = 0.5 \text{ which gives } \theta = \arctan(0.5) = 26.5^\circ$$

$$-0.025 = -0.5(9.8 / (V_0 \cos(26.5^\circ))^2)$$

$$\text{Solve for } V_0 \text{ to obtain } V_0 = 15.6 \text{ m/s}$$

Example 12

A ball is projected from a point O on the edge of a vertical cliff. The horizontal and vertically upward components of the initial velocity are 7 m s^{-1} and 21 m s^{-1} respectively. At time t seconds after projection the ball is at the point (x, y) referred to horizontal and vertically upward axes through O . Air resistance may be neglected.

(i) Express x and y in terms of t , and hence show that $y = 3x - \frac{1}{10}x^2$. [5]

The ball hits the sea at a point which is 25 m below the level of O .

(ii) Find the horizontal distance between the cliff and the point where the ball hits the sea. [3]

(i)	$x = 7t$
	$y = 21t - 4.9t^2$
	$y = 21.x/7 - 4.9 x^2/49$
	$y = 3x - x^2/10$
ii)	$-25 = 3x - x^2/10$ (must be -25)
	solving quadratic
	36.8 m